DSA0602

DATA HANDLING AND VISUALIZATION

CLASS ACTIVITY 2
RELATIONSHIP WORKSHOP IN R

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Course Name	Data Handling and Visualization
Slot	C

Submitted as part of the practical assessment.

General Instructions

The following solutions use R and the packages ggplot2, corrplot, igraph, Rtsne and hexbin. Each question contains code that generates the required chart and prints the numerical results. Run the code in RStudio in the order shown. The random seeds are retained from the question paper.

Note: Numerical results from random datasets should be copied from the R console after running the supplied code. The code prints the exact values for the current R installation; this document does not invent console output.

```
install.packages(c("ggplot2", "corrplot", "igraph", "Rtsne", "hexbin", "tidyr", "viridis"))
```

Q1. Scatterplots – Sales and Football Attendance (5 Marks)

Aim: To examine the relationship between weekly sales and football attendance using scatterplots, correlation and regression models.

R Program

```
set.seed(1217553)
n <- 24
sales <- round(rnorm(n, 85.0, 18.0), 1)
z <- (sales - 85) / 18
football <- round(900 + 200.0 * (0.78 * z + -0.72 * z^2 + 0.40 * rnorm(n)), 0)
football[18] <- round(football[18] + 3.1 * sd(football), 0)
df <- data.frame(sales, football)

# Scatterplot with straight-line and LOESS fits
ggplot(df, aes(x = sales, y = football)) +
  geom_point(size = 2.5, colour = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, colour = "red") +
  geom_smooth(method = "loess", se = FALSE, colour = "darkgreen") +
  labs(title = "Weekly Sales vs Football Attendance",
       x = "Sales (Rs lakh)", y = "Football visitors per day") +
  theme_minimal()

# Correlation and linear regression
cor_value <- cor(df$sales, df$football)
linear_model <- lm(football ~ sales, data = df)
cat("Correlation:", round(cor_value, 3), "\\n")
cat("Linear slope:", round(coef(linear_model)[2], 3), "\\n")
cat("Linear R-squared:", round(summary(linear_model)$r.squared, 3), "\\n")

# Largest standardized residual
std_res <- rstandard(linear_model)
outlier_row <- which.max(abs(std_res))
cat("Largest standardized residual: Row", outlier_row,
    " value =", round(std_res[outlier_row], 3), "\\n")

# Quadratic model and comparison
quadratic_model <- lm(football ~ sales + I(sales^2), data = df)
cat("Quadratic R-squared:", round(summary(quadratic_model)$r.squared, 3), "\\n")
cat("Linear R-squared:", round(summary(linear_model)$r.squared, 3), "\\n")
```

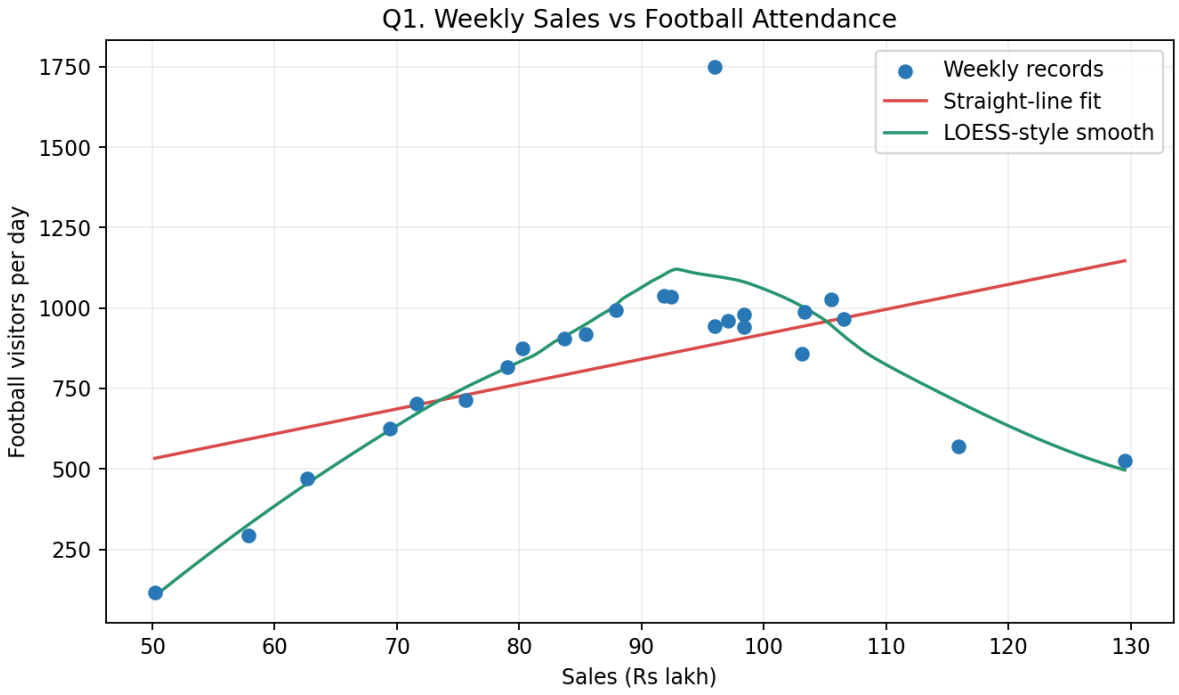
Interpretation / Answers

- The scatterplot displays the association between sales and football visitors. The red line represents the linear fit, while the green curve represents the LOESS fit.

- Use the printed correlation and slope to report direction and strength. A positive slope means visitor count tends to increase as sales increase.
- The row with the largest absolute standardized residual is the most unusual observation relative to the straight-line model. Investigate it rather than deleting it automatically.
- Compare the two R-squared values. The model with the higher R-squared explains more variation in attendance. A visibly curved LOESS line and a higher quadratic R-squared support a non-linear trend.

Recommendation: The Sales Director should use attendance as a useful planning indicator, but should prefer the model with the better fit and investigate unusual weeks before making sales or staffing decisions.

Q1. Scatterplot with linear and smooth trend lines



Output visualization

Q2. Correlogram – Correlation Matrix Heatmap (5 Marks)

Aim: To measure pairwise relationships among six telecom performance indicators and identify strong, weak and negative correlations.

R Program

```
set.seed(1217166)
n <- 30
f1 <- rnorm(n)
f2 <- rnorm(n)
tower_density <- round(14.0 + 4.0 * (0.81 * f1 + 0.58 * rnorm(n)), 1)
drop_rate <- round(2.6 + 0.7 * (0.81 * f1 + 0.59 * rnorm(n)), 1)
traffic_tb <- round(320 + 70 * (0.93 * f2 + 0.38 * rnorm(n)), 0)
users_lakh <- round(24.0 + 6.0 * (0.78 * f2 + 0.63 * rnorm(n)), 1)
churn_pct <- round(rnorm(n, 1.8, 0.5), 1)
arpu <- round(185 + 25 * (0.80 * f1 + 0.60 * rnorm(n)), 0)
df <- data.frame(tower_density, drop_rate, traffic_tb,
                 users_lakh, churn_pct, arpu)

cor_matrix <- cor(df)
print(round(cor_matrix, 3))

# Heatmap centred at zero
cor_long <- as.data.frame(as.table(cor_matrix))
```

```
ggplot(cor_long, aes(Var1, Var2, fill = Freq)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.2f", Freq)), size = 3) +
  scale_fill_gradient2(low = "blue", mid = "white", high = "red",
                       midpoint = 0, limits = c(-1, 1)) +
  labs(title = "Telecom KPI Correlation Heatmap",
       x = NULL, y = NULL, fill = "Correlation") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 35, hjust = 1))

# Strongest pair by absolute correlation, excluding diagonal
diag(cor_matrix) <- 0
strong_pair <- which(abs(cor_matrix) == max(abs(cor_matrix)), arr.ind = TRUE)[1, ]
cat("Strongest pair:", rownames(cor_matrix)[strong_pair[1]], "and",
    colnames(cor_matrix)[strong_pair[2]], "\\n")
cat("Correlation:", round(cor_matrix[strong_pair[1], strong_pair[2]], 3), "\\n")

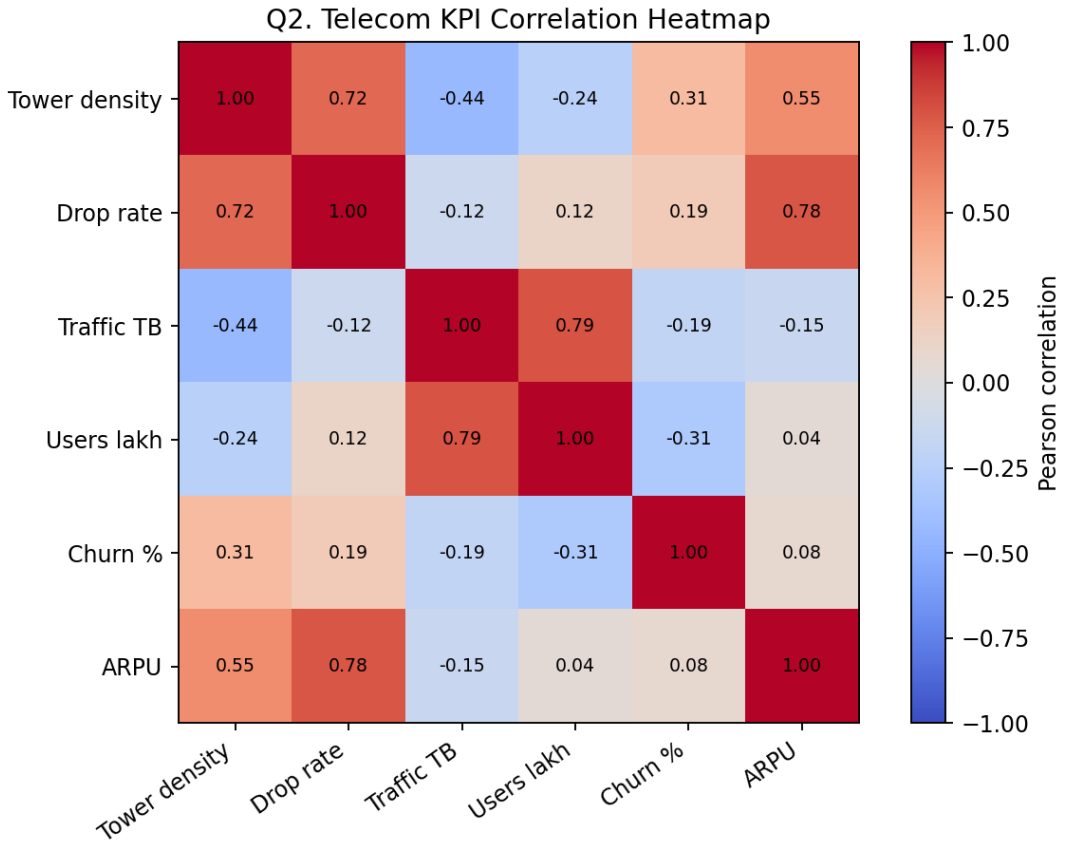
# Weakest pair by absolute correlation
weak_pair <- which(abs(cor_matrix) == min(abs(cor_matrix[upper.tri(cor_matrix)])),
                   arr.ind = TRUE)
# Use the upper triangle to avoid duplicate pairs
upper <- cor_matrix
upper[lower.tri(upper, diag = TRUE)] <- NA
idx <- which(abs(upper) == min(abs(upper), na.rm = TRUE), arr.ind = TRUE)[1, ]
cat("Weakest pair:", rownames(upper)[idx[1]], "and", colnames(upper)[idx[2]], "\\n")
cat("Correlation:", round(upper[idx[1], idx[2]], 3), "\\n")
```

Interpretation / Answers

- The heatmap uses red for positive correlation, blue for negative correlation and white for values near zero.
- The strongest pair is the pair with the largest absolute off-diagonal correlation. Report its variable names and coefficient from the console.
- The weakest pair is the pair with the smallest absolute off-diagonal correlation. A value near zero indicates little linear association.
- Use the matrix to identify any negative pair and explain that the two measures tend to move in opposite directions.

Recommendation: The Network Head should monitor the strongest related KPIs together, while avoiding the assumption that correlation alone proves that one KPI causes another.

Q2. Correlation matrix heatmap



Output visualization

Q3. Dimension Reduction – PCA and t-SNE (5 Marks)

Aim: To reduce six measured variables into fewer dimensions, inspect their variance and compare PCA with a neighbourhood-preserving embedding.

R Program

```
set.seed(1217159)
n <- 36
grp <- rep(c("Group A", "Group C", "Group B"), each = 12)
g <- rep(c(-1, 0, 1), each = 12)
football <- round(120.0 + 30.0 * (rnorm(n) + 0.90 * g), 1)
spend_rs <- round(3200 + 600.0 * (rnorm(n) + 0.90 * g), 0)
stay_nights <- round(2.8 + 0.7 * (rnorm(n) + 0.04 * g), 1)
rating10 <- round(7.6 + 0.8 * (rnorm(n) + 0.04 * g), 1)
occupancy_pct <- round(68.0 + 10.0 * (rnorm(n) + 1.30 * g), 1)
complaints <- round(4.0 + 1.2 * (rnorm(n) + 1.20 * g), 1)
df <- data.frame(football, spend_rs, stay_nights, rating10,
                 occupancy_pct, complaints)

# PCA on scaled variables
pca <- prcomp(df, scale. = TRUE)
variance <- pca$sdev^2 / sum(pca$sdev^2)
cat("PC1 variance (%):", round(variance[1] * 100, 2), "\\n")
cat("PC2 variance (%):", round(variance[2] * 100, 2), "\\n")
cat("Cumulative PC1 + PC2 (%):", round(sum(variance[1:2]) * 100, 2), "\\n")
cat("Components needed for at least 80%:",
    which(cumsum(variance) >= 0.80)[1], "\\n")

# Variable with largest absolute loading on PC1
loadings <- pca$rotation[, 1]
```

```
cat("Largest absolute PC1 loading:", names(which.max(abs(loadings))),
    " loading =", round(loadings[which.max(abs(loadings))], 3), "\\n")

# PCA score plot
scores <- data.frame(PC1 = pca$x[, 1], PC2 = pca$x[, 2], grp = grp)
ggplot(scores, aes(PC1, PC2, colour = grp)) +
  geom_point(size = 2.5) +
  labs(title = "PCA: PC1 vs PC2", x = "PC1", y = "PC2") +
  theme_minimal()

# Group separation using PC1
group_model <- lm(PC1 ~ grp, data = scores)
cat("Group separation R-squared:", round(summary(group_model)$r.squared, 3), "\\n")

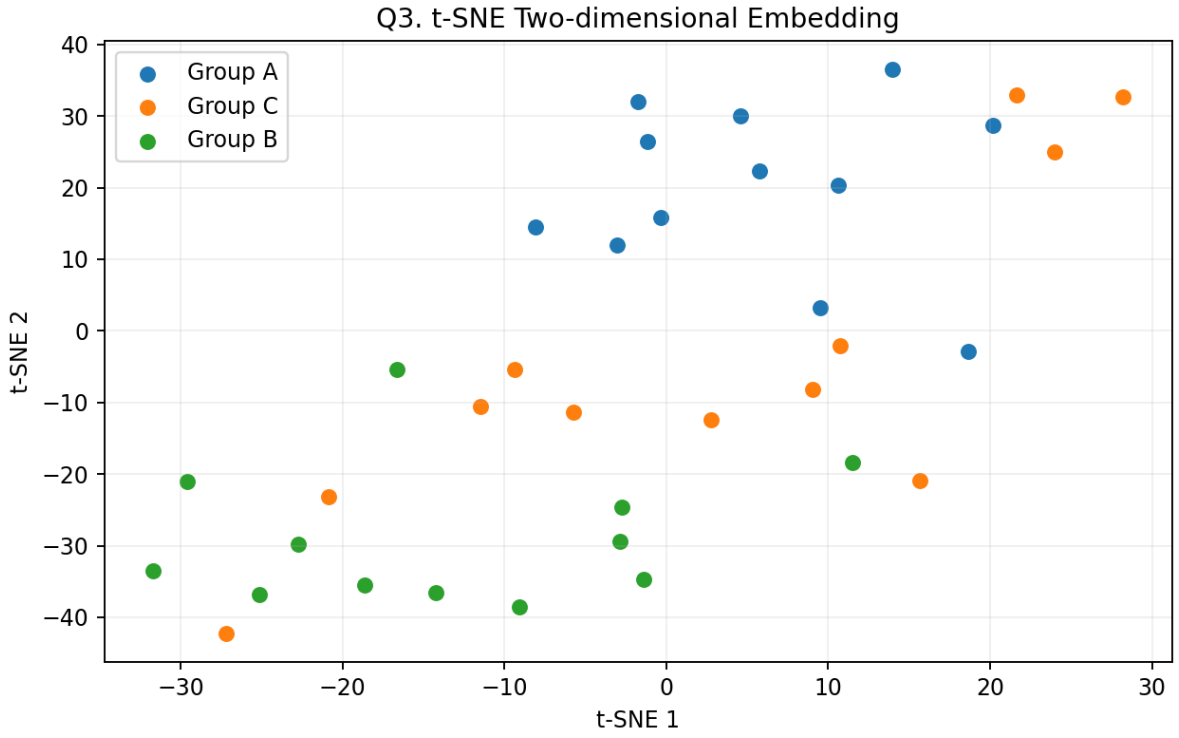
# t-SNE embedding (perplexity must be less than (n - 1) / 3)
tsne <- Rtsne(scale(df), dims = 2, perplexity = 5,
              check_duplicates = FALSE, pca = TRUE)
tsne_df <- data.frame(TSNE1 = tsne$Y[, 1], TSNE2 = tsne$Y[, 2], grp = grp)
ggplot(tsne_df, aes(TSNE1, TSNE2, colour = grp)) +
  geom_point(size = 2.5) +
  labs(title = "t-SNE: Two-dimensional embedding",
       x = "t-SNE 1", y = "t-SNE 2") +
  theme_minimal()
```

Interpretation / Answers

- PC1 and PC2 percentages show how much of the total standardized-data variance is represented by the first two components. The cumulative percentage is their combined contribution.
- The required component count is the first component at which cumulative explained variance reaches 80%.
- The variable with the largest absolute PC1 loading contributes most strongly to PC1; the sign indicates the direction of its contribution.
- PCA is a linear method that preserves overall variance. t-SNE focuses on local neighbourhoods and can reveal clusters, but its axes do not represent explained variance and global distances should not be over-interpreted.

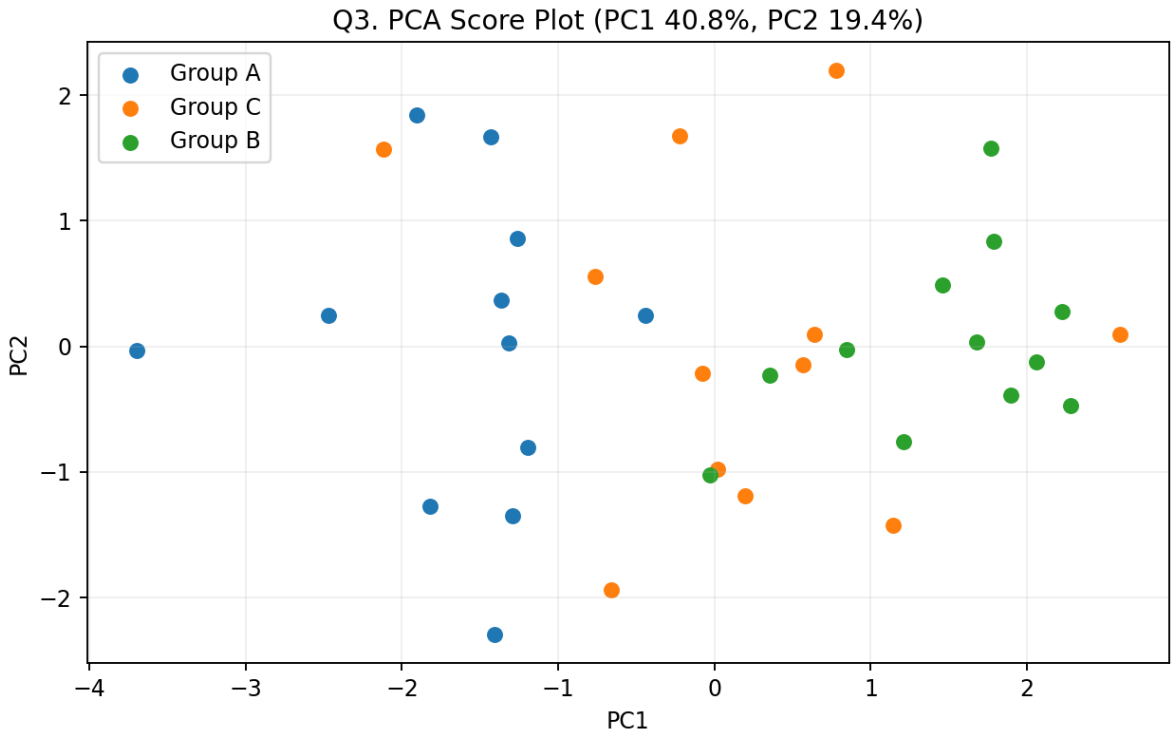
Recommendation: Use PCA for compact reporting and variance explanation. Use t-SNE as an exploratory visualization when local group structure is important.

Q3. t-SNE embedding



Output visualization

Q3. PCA score plot



Output visualization

Q4. Network Graph – Parcels Delivered (5 Marks)

Aim: To build a weighted network of six hubs, calculate centrality and network density, and examine the effect of removing one connection.

R Program

```
set.seed(1217212)
nodes <- c("Kolkata", "Mumbai", "Ahmedabad", "Chennai", "Hyderabad", "Delhi")
pairs <- t(combn(nodes, 2))
u <- runif(nrow(pairs))
keep <- (u < 0.31) | ((pairs[, 1] == "Delhi" | pairs[, 2] == "Delhi") & u < 0.95)
edges <- data.frame(from = pairs[keep, 1],
                    to = pairs[keep, 2],
                    weight = round(runif(sum(keep), 1, 26), 0))

g <- graph_from_data_frame(edges, directed = FALSE, vertices = nodes)

# Weighted network plot
plot(g, vertex.size = 30, vertex.label.color = "black",
     edge.width = E(g)$weight / 5,
     main = "Parcel Delivery Hub Network")

# Centrality
deg <- degree(g)
btw <- betweenness(g, directed = FALSE, weights = NA)
centrality <- data.frame(Node = names(deg), Degree = as.numeric(deg),
                         Betweenness = as.numeric(btw))
print(centrality)
central_node <- names(which.max(btw))
cat("Most central node by betweenness:", central_node, "\\n")
```

```
# Network summary
cat("Number of edges:", ecount(g), "\\n")
cat("Network density:", round(edge_density(g), 3), "\\n")
cat("Connected components:", components(g)$no, "\\n")

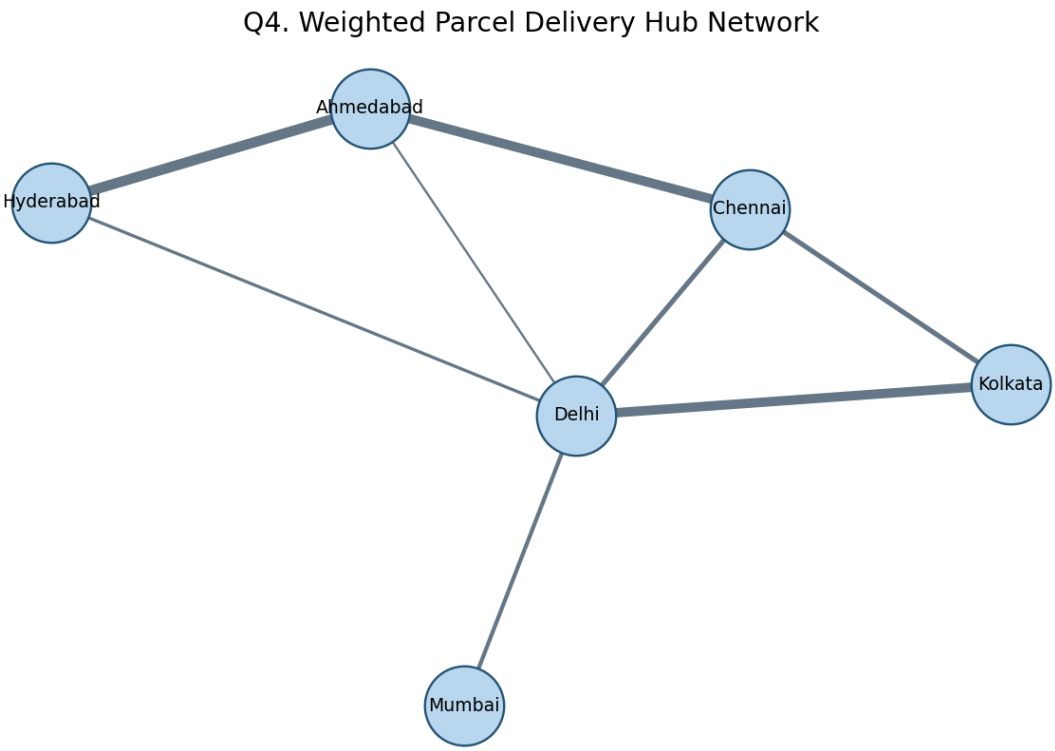
# Remove the weakest-weight edge and recalculate components
weak_edge <- which.min(E(g)$weight)
g2 <- delete_edges(g, weak_edge)
cat("After deleting weakest edge:\\n")
cat("Edges:", ecount(g2), "\\n")
cat("Connected components:", components(g2)$no, "\\n")
```

Interpretation / Answers

- Node size represents connectivity and edge width represents parcel-link weight. A larger degree means a hub has more direct connections.
- Betweenness measures how often a node lies on shortest paths between other nodes. The node with the highest betweenness is the most important bridge in this network.
- Network density is the proportion of possible edges that actually exist. The connected-component count indicates how many separate groups can communicate through the network.
- After deleting the weakest edge, compare the component count. If it increases, that link was important for keeping parts of the network connected.

Recommendation: The Operations Head should protect high-betweenness hubs and critical links with backup routes, since disruption at these points can affect many parcel paths.

Q4. Weighted network graph



Output visualization

Q5. Paired Data – Before and After Process Change (5 Marks)

Aim: To compare waiting times for the same eight departments before and after a process change using a paired slope chart and paired t-test.

R Program

```
set.seed(1217323)
entity <- c("Cardiology", "Oncology", "Pharmacy", "Orthopaedics",
            "Paediatrics", "Neurology", "Radiology", "Emergency")
n <- 8
before <- round(rnorm(n, 42.0, 5.0), 1)
change <- round(c(-1, 1, -1, -1, 1, 1, 1, 1) * runif(n, 0.4, 1.4) * 6.0, 1)
after <- before + change
df <- data.frame(entity, before, after)

# Paired slope chart
library(tidyr)
long <- pivot_longer(df, cols = c(before, after),
                     names_to = "period", values_to = "wait_time")
ggplot(long, aes(period, wait_time, group = entity, colour = entity)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2.5) +
  labs(title = "Department Waiting Time: Before vs After",
       x = "Period", y = "Waiting time (minutes)") +
  theme_minimal()

# Classify changes using the specified 1% threshold
threshold <- 0.01 * mean(before)
change_value <- after - before
status <- ifelse(abs(change_value) < threshold, "Flat",
                 ifelse(change_value < 0, "Improved", "Worsened"))
result <- data.frame(entity, before, after, change_value, status)
print(result)
cat("Flat threshold:", round(threshold, 3), "minutes\\n")

# Largest change
largest <- which.max(abs(change_value))
cat("Largest change:", entity[largest],
    " amount =", round(change_value[largest], 1),
    " percentage =", round(change_value[largest] / before[largest] * 100, 2), "%\\n")

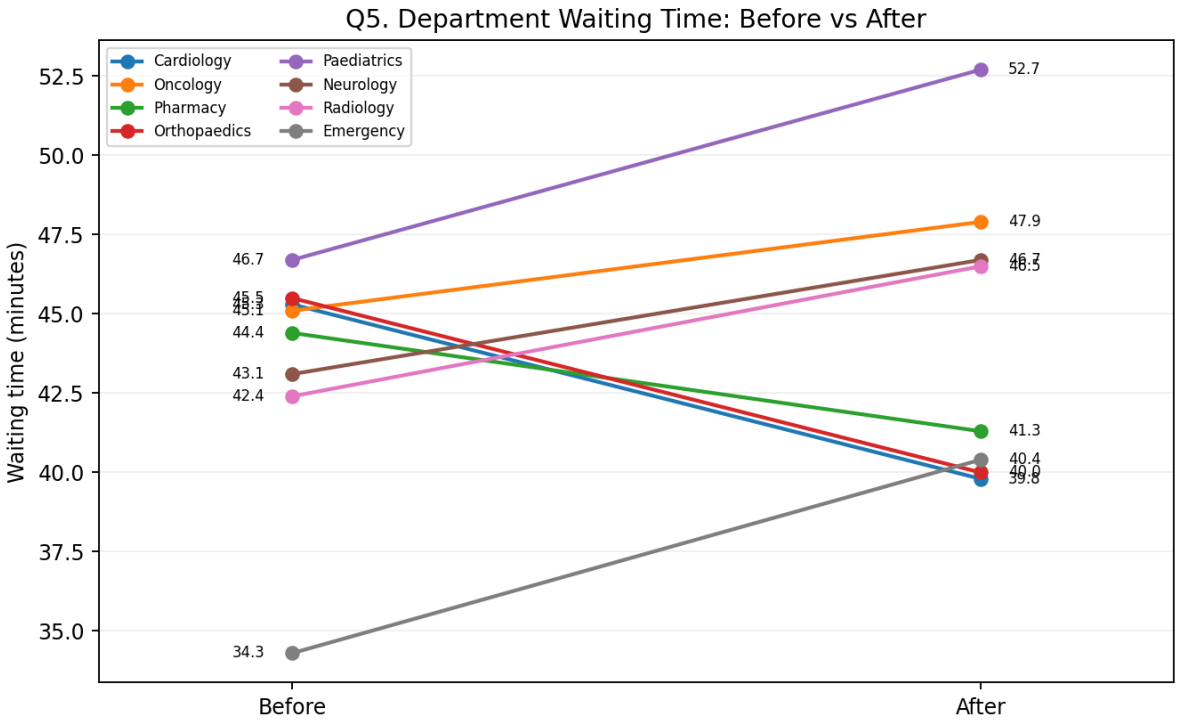
# Paired t-test and mean change
test <- t.test(after, before, paired = TRUE)
print(test)
cat("Mean change (after - before):", round(mean(change_value), 2), "minutes\\n")
```

Interpretation / Answers

- A downward line means waiting time decreased and the department improved. An upward line means waiting time increased and the department worsened.
- A change is classified as flat when its absolute value is below 1% of the mean before waiting time. Otherwise, a negative change is improved and a positive change is worsened.
- The largest change is the department with the greatest absolute difference. The percentage change is (after − before) divided by before, multiplied by 100.
- The paired t-test checks whether the mean within-department change differs significantly from zero. Interpret its p-value together with the mean change and confidence interval.

Recommendation: Continue the process change if the average waiting time has fallen meaningfully and the paired test supports an improvement. Review departments whose waiting time increased.

Q5. Paired slope chart



Output visualization

Q6. Bubble Chart – Marketing Spend, Watch Time and Subscribers (5 Marks)

Aim: To compare six content genres using marketing spend, watch time and subscriber count.

R Program

```
set.seed(1217318)
genre <- c("Drama", "Thriller", "Documentary", "Animation", "Sports", "Comedy")
n <- 6
z <- rnorm(n)
spend_lakh <- round(45.0 + 12.0 * z, 1)
watch_hrs <- round(11.0 + 3.0 * (0.90 * z + 0.40 * rnorm(n)), 1)
subs_lakh <- round(18.0 + 4.0 * rnorm(n), 1)
subs_lakh[5] <- round(18.0 * 2.4, 1)
spend_lakh[5] <- round(45.0)
watch_hrs[5] <- round(11.0)
df <- data.frame(genre, spend_lakh, watch_hrs, subs_lakh)

# Bubble chart
ggplot(df, aes(spend_lakh, watch_hrs, size = subs_lakh, label = genre)) +
  geom_point(alpha = 0.65, colour = "steelblue") +
  geom_text(vjust = -0.8, size = 3) +
  scale_size_area(max_size = 18, name = "Subscribers (lakh)") +
  labs(title = "Marketing Spend vs Watch Time by Genre",
       x = "Marketing spend (Rs lakh)",
       y = "Watch time (hours per user per month)") +
  theme_minimal()

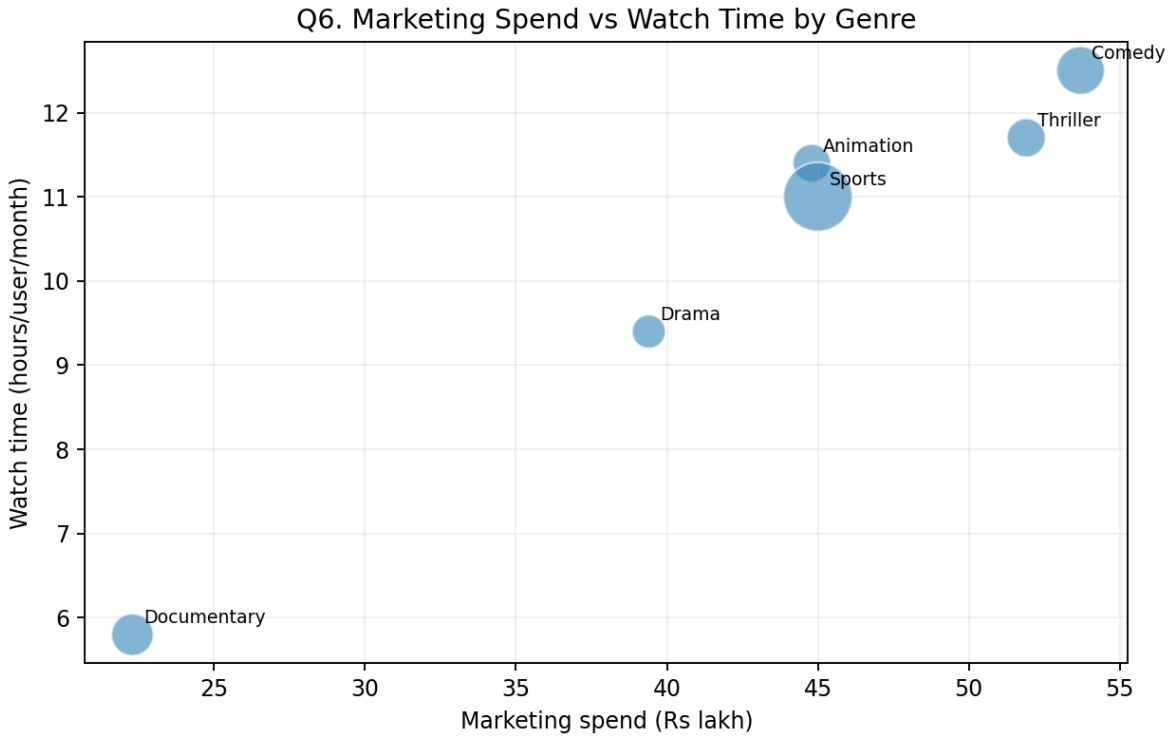
# Correlation and z-score outlier check
cat("Correlation:", round(cor(spend_lakh, watch_hrs), 3), "\\n")
z_subs <- as.numeric(scale(subs_lakh))
outlier <- which.max(abs(z_subs))
cat("Largest subscriber z-score:", genre[outlier],
    " z =", round(z_subs[outlier], 3), "\\n")
cat("Outlier is Sports:", genre[outlier] == "Sports", "\\n")
```

Interpretation / Answers

- The x-axis shows marketing spend, the y-axis shows monthly watch time, and bubble area represents subscribers. Larger bubbles indicate more subscribers.
- The correlation coefficient reports the direction and strength of the linear relationship between spend and watch time.
- The subscriber outlier is the genre with the largest absolute z-score. Sports is deliberately assigned an unusually high subscriber count in the supplied data.
- A plain scatterplot would hide subscriber volume because it only displays x and y positions. The bubble size adds a third variable.

Recommendation: The Content Head should consider both viewing time and subscriber volume. A genre with many subscribers may deserve attention even when its spend and watch-time coordinates are not extreme.

Q6. Bubble chart



Output visualization

Q7. Hexbin Plot – Loan Amount and Applicant Income (5 Marks)

Aim: To visualize 3,000 loan applicants, identify the busiest region and compare the number of distinct coordinate pairs with the information shown by a scatterplot.

R Program

```
set.seed(1217371)
n <- 3000
core <- runif(n) < 0.85
loan_lakh <- round(ifelse(core, rnorm(n, 38.0, 5.4),
                          rnorm(n, 11.0, 65.0)), 1)
income_k <- round(ifelse(core, 52.0 + 0.59 * (loan_lakh - 38.0) +
                         7.2 * rnorm(n), runif(n, 16.0, 88.0)), 1)
df <- data.frame(loan_lakh, income_k)

# Hexbin plot with count legend
ggplot(df, aes(loan_lakh, income_k)) +
  geom_hex(bins = 25) +
```

```
  scale_fill_viridis_c(name = "Applicant count") +
  labs(title = "Loan Amount vs Applicant Income",
       x = "Loan amount (Rs lakh)",
       y = "Applicant income (Rs thousand per month)") +
  theme_minimal()

# 10 x 10 grid counts
x_breaks <- seq(min(df$loan_lakh), max(df$loan_lakh), length.out = 11)
y_breaks <- seq(min(df$income_k), max(df$income_k), length.out = 11)
grid <- table(cut(df$loan_lakh, breaks = x_breaks, include.lowest = TRUE),
              cut(df$income_k, breaks = y_breaks, include.lowest = TRUE))
cell <- which(grid == max(grid), arr.ind = TRUE)[1, ]
cat("Busiest grid cell x range:",
    levels(cut(df$loan_lakh, breaks = x_breaks, include.lowest = TRUE))[cell[1]], "\\n")
cat("Busiest grid cell y range:",
    levels(cut(df$income_k, breaks = y_breaks, include.lowest = TRUE))[cell[2]], "\\n")
cat("Busiest grid count:", max(grid), "\\n")
cat("Share of all applicants:", round(max(grid) / n * 100, 2), "%\\n")

# Distinct coordinate pairs
distinct_pairs <- nrow(unique(df))
cat("Distinct (loan, income) pairs:", distinct_pairs, "out of", n, "\\n")
cat("Duplicate coordinate pairs:", n - distinct_pairs, "\\n")
```

Interpretation / Answers

- The hexbin colour scale represents the number of applicants in each hexagonal region. Darker or more intense cells indicate higher concentration, according to the legend.
- The busiest 10 × 10 grid cell is reported by its loan-amount range, income range, count and percentage of all 3,000 applicants.
- The distinct-pair count is the number of unique loan-amount and income combinations. It can be lower than 3,000 because values are rounded to one decimal place, creating duplicate coordinates.
- A scatterplot can suffer from overplotting when many observations share or nearly share positions. A hexbin plot groups nearby observations and communicates density more clearly.

Recommendation: The Finance Manager should use the hexbin plot to identify the most common applicant profile and investigate dense or unusual regions before setting loan review policies.

Overall Conclusion

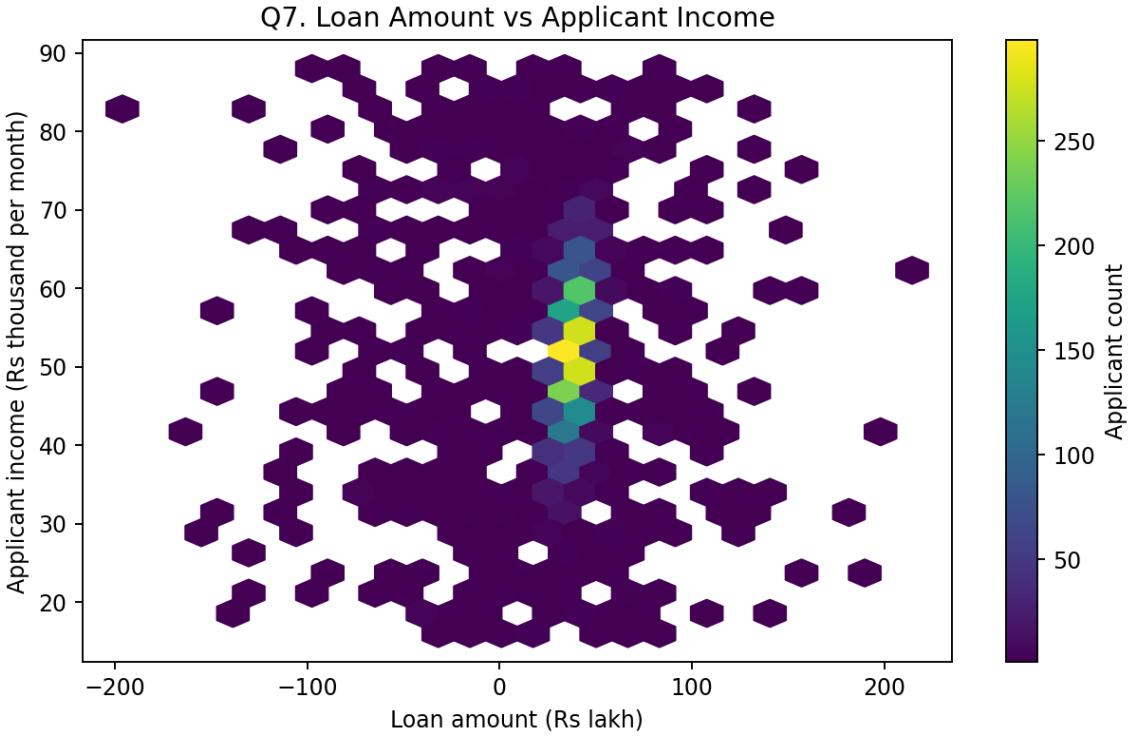
This activity applies seven relationship-analysis techniques in R: scatterplots and regression, correlation heatmaps, PCA and t-SNE, network graphs, paired comparisons, bubble charts and hexbin plots. Together, these methods help identify trends, associations, clusters, influential network nodes, process changes, outliers and dense regions in data. Numerical findings should be verified from the R console after executing the supplied programs with the specified random seeds.

Code of Conduct

I certify that this work is prepared for my class activity and that I will verify the program outputs by running the code in R.

Signature: _____

Q7. Hexbin density plot



Output visualization